

Cleaved hemagglutinin in ESM3:

Can the model treat HA1 and HA2 as one molecule
joined by a disulfide bridge?

my_extendedSampling / softmax / bridge_test

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Abstract

`zero_polymer` is an influenza hemagglutinin precursor (HA0). In the mature protein it is cut once, after Arg329 in mature numbering, into HA1 and HA2, which stay together through an interchain disulfide between Cys14 of HA1 and Cys137 of HA2. The question was whether ESM3 can be given that two-chain molecule and handle it. The mechanical answer is yes: the stock tokenizer accepts the chain-break character, but no benefit of declaring the break shows up consistently across decoding step counts. The structural answer is no, with caveats: no assembled construct folds well enough to pass the geometry gate, and the interchain Cys14–Cys137 distance is 96 Å (cleaved) and 130 Å (no break), against a bonding criterion of 4.5 Å. Experimental structures of the two disulfide controls then showed that the 4.5 Å criterion is exact on real geometry: the native bonds, and nothing else, lie within it. The failures are in the predictions. The predicted crambin fold is wrong (TM-score 0.32) at pLDDT 83 and passes the geometry gate, and the model places a non-native pair at 3.3 Å that is 17.9 Å in the experiment. The round trip reproduces both folds to about 0.3 Å, so the representation can hold the bonds; a coordinate prompt pulls a pair partway toward native geometry without holding it. This report records the results, the corrections made along the way, the limitations, and a prioritised to-do list.

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1 Summary

- **Mechanics work.** `model.encode(ESMProtein(sequence=...))` accepts |; 551 positions go in and 551 come out; `to_protein_complex()` recovers two chains. No change to the model code was needed.
- **No consistent benefit of the chain-break.** At 256 steps HA1's mean pLDDT is 51.4 fused to HA2, 63.2 with the break and 65.2 alone, which looked like a benefit. Across 64, 128 and 256 steps the break-minus-fused difference is -5.2 , $+2.4$, $+11.8$: the sign is not stable, and the earlier claims that the break helps HA1 and its Cys14 anchor are withdrawn.
- **Step count is a large, non-monotone effect.** HA1 alone scores 48.4, 67.0, 65.2 and 59.8 at 64, 128, 256 and 329 steps. Each step count is a deterministic but effectively independent draw, and draws differ by more than most effects claimed so far. Only sign-stable differences are findings.
- **HA2 alone folds well; every assembly degrades it.** HA2 alone passes the gate at all three step counts (pLDDT 88–94, no clashes) and is 17–29 points more confident than HA2 in any assembly. No assembled construct, and no HA1-alone run, passes the gate at any step count.
- **The interchain bridge is not formed** in either assembled construct, by a margin (at least $10\times$ the criterion at every step count) that no threshold subtlety can change.
- **The $4.5\text{\AA}$ criterion is exact on experimental structures.** In BPTI and crambin, exactly the native bonds lie within $4.5\text{\AA}$ and nothing else does. The earlier verdict that it lacks specificity was wrong: the non-native pair is a prediction error.
- **The predicted crambin fold is wrong** (TM-score 0.32), at pLDDT 83 and with no clashes. The geometry gate is necessary, not sufficient.
- **The representation can hold the bonds.** The round trip reproduces both folds to about $0.3\text{\AA}$. A coordinate prompt pulls a bridge partway toward native geometry, locally, at a cost in confidence, and does not hold it.
- **Not yet tested:** the controls at other step counts, the trimer, an experimental HA structure, partner identity for HA.

2 What was built

All of it is on branch `claude/esm3-disulfide-polymer-chains-rwvpoq`.

Commit	Content
46bccd8	<code>tests/references.py</code> : chains, bridge, numbering helpers
1f35fbf, fcb86aa, caeb436	<code>tests/test_cleaved_complex.py</code> , three revisions
4056d91, 346b3de	BPTI and crambin controls; known-disulfide measurement
086e2cd, 1b0d3ce, 6b34313,	<code>bridge_test/DEVLOG_bridge_test.txt</code>
8efcd2c	
87dc571	<code>bridge_test/DEVLOG_calibration.txt</code>

The construct. Chain one is mature HA1 (329 aa, ends ...QTR); chain two is HA2 (221 aa, starts GIFGAI); the 16-residue signal peptide is dropped, so 550 residues in total. Both chains are slices of ZERO_POLYMER, never retyped, and `_check()` asserts the junction, the lengths, and that both anchors are cysteines.

A numbering trap worth keeping. The HA landmarks are in *mature* numbering, which runs 16 behind the index into ZERO_POLYMER. Read naively, ZERO_POLYMER[328] ("Arg329") is T and ZERO_POLYMER[13] ("Cys14") is V. Neither raises. `mature_to_index()` and `chain_index()` exist so the conversion is never done by hand.

3 Method

Route into the model. The sampler’s usual path (`ExtendedProtein` with `predict_attention` on soft `sequence_probs`) cannot carry a chain break: `expand()` maps through a 25-letter vocabulary with no |, and the encoder zero-pads over token 31. The stock `encode/predict_protein` route runs the real tokenizer and has none of these problems. Two shortcuts that look right are wrong: `chain_id=` is consumed only by geometric attention, which the soft path masks and zeroes, so it is a silent no-op; and `sequence_id=` is an attention separator, so distinct ids would make the chains ignore each other.

What the decoder returns. Backbone plus an inferred $C\beta$, and no side-chain sulfur at any residue. So SG-SG is unavailable and $C\beta$ - $C\beta$ distance is the only disulfide proxy. $C\beta$ is not predicted: `infer_cbeta()` derives it from N/CA/C and writes NaN at every glycine. The chain-break position is excluded from all $C\beta$ statistics.

The instrument.

- *Geometry gate:* $pTM \geq 0.40$, and no $C\beta$ - $C\beta$ pair closer than $3.0 \text{ \AA}$ at sequence separation ≥ 2 .
- *Bond criterion:* $C\beta$ - $C\beta \leq 4.5 \text{ \AA}$.
- *Controls:* protein G (fold control, no cysteines); BPTI and crambin (three known disulfides each).

4 Results

4.1 The controls

Reference	steps	pTM	pLDDT	% ≥ 70	min $C\beta$ ($\text{\AA}$)	clashes
protein G (56 aa)	56	0.7596	93.24	100.0	3.68	0
BPTI (58 aa)	58	0.7545	92.89	98.3	3.35	0
crambin (46 aa)	46	0.5816	83.31	89.1	3.33	0

All three pass the gate, so the harness folds, and the choice of $3.0 \text{ \AA}$ as the clash line is borne out empirically (confident folds have their closest pair at 3.3 – $3.7 \text{ \AA}$).

4.2 HA constructs

	cleaved	mature	HA1 alone	HA1 alone	HA2 alone
positions / steps	551 / 256	550 / 256	329 / 256	329 / 329	221 / 221*
pTM	0.2322	0.2614	0.4642	0.4133	0.5674
pLDDT, all residues	63.89	59.20	65.16	59.83	93.93
% residues ≥ 70	45.5	31.8	37.1	18.5	97.7
pLDDT, HA1 (% ≥ 70)	63.22 (38.6)	51.42 (11.2)	65.16 (37.1)	59.83 (18.5)	—
pLDDT, HA2 (% ≥ 70)	64.90 (55.7)	70.78 (62.4)	—	—	93.93 (97.7)
min $C\beta$ – $C\beta$ (Å)	1.50	1.17	2.01	0.44	4.14
clashes	15	9	7	16	0
gate	fail	fail	fail	fail	pass
Cys14–Cys137 $C\beta$ (Å)	95.67	129.74	—	—	—
pLDDT at Cys14 / Cys137	73.6 / 61.3	22.5 / 80.9	—	—	—

The first run, at one decoding step, gave pTM 0.1827 and a bridge distance of 38.55 Å with a 1.64 Å closest cysteine pair. It is not tabulated because it measured a structure the model did not believe in; see Section 5. *HA2 alone was requested at 256 steps and clamped to 221 by the script (a chain cannot have more steps than positions), so it has been seen at one step count only.

Reproducibility. Cleaved and mature were each run twice, with different script versions, on the same machine, and gave identical pTM (0.2322 and 0.2614) and identical bridge distances (95.67 and 129.74 Å). HA2 alone was run twice as well (the second time by the clamp above) with identical output to every printed digit. Decoding at temperature zero looks deterministic here. This is three repeat pairs on one machine, not a proof.

4.3 Step sweep

	64 steps	128 steps	256 steps	
pLDDT, HA1 in cleaved	56.16	64.32	63.22	
pLDDT, HA1 fused	61.35	61.90	51.42	
pLDDT, HA1 alone	48.43	67.01	65.16	
pLDDT, HA2 in cleaved	68.49	63.27	64.90	
pLDDT, HA2 fused	68.94	71.40	70.78	
pLDDT, HA2 alone*	93.67	88.47	93.93	
pLDDT at Cys14, cleaved	31.7	58.0	73.6	
pLDDT at Cys14, fused	41.1	69.9	22.5	
clashes, cleaved / fused	22 / 9	19 / 2	15 / 9	
clashes, HA1 alone / HA2 alone	17 / 0	5 / 0	7 / 0	
bridge $C\beta$ – $C\beta$, cleaved (Å)	98.8	137.9	95.7	
bridge $C\beta$ – $C\beta$, fused (Å)	164.3	48.0	129.7	
Contrast at equal steps	64	128	256	sign
HA1 pLDDT: break – fused	–5.2	+2.4	+11.8	mixed
HA1 pLDDT: alone – cleaved	–7.7	+2.7	+1.9	mixed
Cys14 pLDDT: break – fused	–9.4	–11.9	+51.1	mixed
HA2 pLDDT: alone – cleaved	+25.2	+25.2	+29.0	+ at all
HA2 pLDDT: alone – fused	+24.7	+17.1	+23.2	+ at all
HA2 pLDDT: break – fused	–0.5	–8.1	–5.9	– at all

*HA2 alone is capped at its 221 residues, so its 256-step column is the 221-step run. Computed by `bridge_test/sweep_contrasts.py`; $n = 3$, so means and spreads are descriptive and no p-value is claimed.

4.4 Findings

The chain-break’s benefit to HA1 is a 256-step effect. Break minus fused is -5.2 , $+2.4$, $+11.8$ at 64, 128, 256 steps. The $+11.8$ comes from fused HA1 dipping to 51.42 at 256 steps (61.35 and 61.90 at the other two), while cleaved HA1 stays at 56–64. The Cys14 anchor result is the same artefact: at 64 and 128 steps the break gives the *lower* anchor confidence. Neither claim passes the sign-at-every-step rule.

The clash *topology* differs. At 256 steps mature has fewer clashes (9 against 15) but they sit at $|i - j|$ of 80–159, so distant segments are superimposed; cleaved’s are local ($|i - j| \leq 16$). Across the sweep this holds at two of three step counts (fused is long-range at 128 and 256 but not at 64; cleaved has one long-range clash, at 182, at 128 steps), so it is suggestive, not clean.

HA2 is far more confident alone than in any assembly. HA2 alone minus HA2 in the cleaved complex is $+25.2$, $+25.2$, $+29.0$, and against the fused one $+24.7$, $+17.1$, $+23.2$: the same sign at every step count, with spreads of 2–4 points on differences of 17–29. This is the one robust contrast in the study. The break costs HA2 a little (-0.5 , -8.1 , -5.9 , negative throughout, but of the same size as the step noise).

HA1 alone against HA1 in the complex is undetermined: -7.7 , $+2.7$, $+1.9$. At 64 steps HA1 alone is unfolded (0.0% of residues above 70). Whether the complex is right in a way pLDDT cannot see, for example in the interchain placement, is untested; the assembled pTM is too low to resolve it.

Step count is a large, non-monotone effect. HA1 alone scores 48.43, 67.01, 65.16 and 59.83 at 64, 128, 256 and 329 steps; it rises, plateaus and falls. Fused HA1 drops 10 points between 128 and 256 steps while cleaved HA1 does not. Which cysteines pair also moves with steps alone (one:64–76 goes from 3.92 Å to 26.64 Å between 329 and 256 steps).

HA1 is the hard half in every construct. Alone it fails the gate at both step counts tried: 16 clashes at 329 steps (minimum C β separation 0.44 Å) and 7 at 256 steps (2.01 Å); it fails at 64 and 128 too.

The bridge is not formed, at any step count. Cleaved gives 98.8, 137.9, 95.7 Å and fused 164.3, 48.0, 129.7 Å, against a 4.5 Å criterion. No assembled run passes the gate either (cleaved 22, 19, 15 clashes; fused 9, 2, 9, the 2-clash run failing on pTM), and HA1 alone fails at every step count tried.

4.5 Against experimental structures

Three experimental entries were fetched, and each was accepted only if its own sequence matched the reference: BPTI (5PTI, 1.00 Å), crambin (3NIR, 0.48 Å) and protein G (1PGB, 1.92 Å). C β is taken by the model’s own convention on both sides; the deposited C β differs by about 0.1–0.2 Å.

	protein G	BPTI	crambin
CA RMSD (Å)	2.96	1.22	7.91
TM-score (approx.)	0.835	0.922	0.316
contact F1, $ i - j \geq 6$	0.80	0.92	0.35
local CA–CA error, all pairs (Å)	1.10	0.41	4.06
worst 7-residue window (Å)	2.68 (11–17)	0.78 (52–58)	3.41 (12–18)
pTM / mean pLDDT (model)	0.76 / 93	0.75 / 93	0.58 / 83

The crambin fold is wrong, and passes the gate. A TM-score of 0.32 is well below the 0.5 that marks a shared fold, and the local error is 4 Å on average. The model gives it pTM 0.58 and pLDDT 83, with no clashes. The geometry gate is necessary, not sufficient.

Protein G’s fold is right, and one loop is confidently wrong. Residues 9–15 lie 5–11 Å from experiment at pLDDT 80–91. Their internal shape is wrong too: the intra-loop CA–CA error averages 1.8 Å, and the end-to-end distance is 15.3 Å against 11.3 Å. Their experimental B-factors are normal, so this is not crystal disorder, and confidence does not flag the worst residues.

The 4.5 Å criterion is exact on experimental geometry. All six supplied pairings are disulfides in the files (SG–SG 2.02–2.05 Å). Exactly the native bonds lie within 4.5 Å in both proteins. The nearest non-bonded pair is 6.14 Å in BPTI and 5.79 Å in crambin. The earlier claim that the criterion lacks specificity was wrong: the non-native crambin pair Cys16–Cys40 is predicted at 3.33 Å and is 17.9 Å in the experiment. That specificity failure belongs to the prediction.

Protein	Pair	SG–SG (Å)	exp. Cβ	pred. Cβ	pred. within 4.5 Å
BPTI	Cys5–Cys55	2.04	3.49	3.66	yes
BPTI	Cys14–Cys38	2.03	3.89	3.61	yes
BPTI	Cys30–Cys51	2.02	3.71	4.45	yes
crambin	Cys3–Cys40	2.03	3.45	7.75	no
crambin	Cys4–Cys32	2.05	4.03	3.79	yes
crambin	Cys16–Cys26	2.04	3.82	10.42	no
crambin	Cys16–Cys40 (non-native)	–	17.88	3.33	yes

The round trip holds both folds. Experimental coordinates encoded to structure tokens and decoded reproduce the native fold to 0.31 Å (BPTI) and 0.30 Å (crambin), with TM-scores 0.99 and 0.98. The known bonds come back within about 0.4 Å. The representation can hold these folds and these bonds. For crambin the failure is in generation, not in the representation.

BPTI cannot test imposition. Its bonds form without any prompt. A prompt on Cys14 and Cys38 leaves all three bond distances within 0.02 Å of the unprompted values.

A coordinate prompt pulls crambin’s pair partway, and costs confidence. Prompting the backbone of residues 3 and 40 moves their CA–CA distance from 7.84 to 6.68 Å (experiment 5.66 Å; 53% of the gap closed) and their Cβ–Cβ distance from 7.75 to 5.14 Å (experiment 3.45 Å; 61% closed). It forms no other bond, the fold stays wrong (RMSD 7.9 to 7.1 Å), and pTM falls from 0.58 to 0.46. This is a soft, partial, local pull, not an imposition. The crambin fold is wrong, so the test is confounded either way.

5 Corrections made during the session

These are recorded because several first readings were wrong.

- One decoding step.** The first verdict (“bridge not formed, 38.55 Å”) came from a script that used the default `num_steps=1`, which samples all 551 structure tokens at once. It scored pTM 0.18 with a 1.64 Å “contact” shorter than a C–C bond. The verdict was worthless. The gate now runs before any verdict.
- A control that did not control.** Protein G validated the fold but has no cysteines, so it could not validate the measurement. BPTI and crambin were added for that reason.

3. **Two overstated claims, retracted.** The calibration script printed “4.5 Å is sound; HA2’s disulfides are not formed” from BPTI alone. That tested sensitivity on one protein and specificity not at all. Separately, HA1-alone pairs at 3.51 and 3.92 Å had been described as “real disulfide geometry”; by the crambin result they could be non-native.
4. **A pre-registered prediction failed.** I expected the trimer signature (both chains confident, assembly low). The result was asymmetric instead: HA2 adequate, HA1 poor.
5. **“HA2’s presence helps HA1” retracted.** It rested on HA1 alone at 329 steps against HA1 in the complex at 256. At matched steps HA1 alone is 65.16 against 63.22. This report had listed that confound as its first limitation, and the rerun resolved it against the claim. I had also said the premise “ESM3 should not treat the chains as separate” was confirmed; what is confirmed is narrower (an unbroken fusion is wrong, and the break repairs it).
6. **Three more findings withdrawn after the step sweep.** The chain-break benefit to HA1, its benefit to the Cys14 anchor, and the narrower “break repairs fusion” that replaced the first claim all rested on 256 steps. I stated each with more confidence than one step count could support. The sweep also exposed a bug in my own contrast script, which kept an old pre-pLDDT result file for the 256-step runs; an empty column gave it away.
7. **Two calibration verdicts, retracted.** The crambin calibration called the 4.5 Å criterion unreliable, but it was computed on a misfolded protein and says nothing about the criterion. The claim that the criterion lacks specificity was also wrong: experimental geometry shows it exact, and the non-native crambin pair is a model error. The supplied disulfide pairings are now confirmed as bonds in the experimental files.
8. **Small bugs.** The chain-break position was admitted into the $C\beta$ statistics; labels were wrong for the break-free reference; a results file was overwritten by a generic filename.

6 Limitations

6.1 Of the measurements

- **Experimental ground truth exists only for the controls.** There is no experimental HA structure, so nothing about HA’s fold or bridge has been compared with experiment. Each control was predicted at one step count.
- **$C\beta$ – $C\beta$ is a proxy.** The decoder emits no sulfur, and $C\beta$ is inferred from the backbone. A bond criterion built on it inherits any backbone error, and glycines (which have no $C\beta$) are invisible to the clash count.
- **Three step counts per construct.** Enough to show that the sign of several contrasts flips, not enough to characterise the step dependence. HA2 alone is capped at 221 steps. It is unknown whether steps or tokens decoded per step is the operative variable.
- **One run per condition.** The three repeat pairs agree exactly, but no condition has been run across seeds or checkpoints, and nothing here gives error bars.
- **pTM depends on length,** so comparing it between a 46-residue and a 550-residue chain is weak. Only same-construct comparisons are clean.
- **Two controls.** BPTI is small and folds correctly, so it cannot test imposition. Crambin’s fold is wrong, so its results are confounded. Neither says what the criterion does in a 550-residue fold.

- **The control pairings are confirmed by experiment.** Each supplied pairing is a disulfide in the deposited file (SG-SG 2.02–2.05 Å). The sequences were checked against the experimental SEQRES, not against UniProt.
- **Only esm3-open.** No other checkpoint was tried.

6.2 Of the scope

- **Monomer only.** HA is a trimer. A single HA1+HA2 protomer may have no self-contained fold, and a low score may then be correct. Not tested.
- **No verified native disulfide list for HA,** so partner recovery cannot be run on HA at all. Positive bridge claims about HA are not possible yet.
- **The energy machinery cannot take a chain break.** The sampler’s soft-probability path has no way to carry token 31, so none of the U_{am} work has been applied to the cleaved form.
- **Glycans, the fusion peptide, and everything outside the sequence** are absent by construction.

6.3 Of the session itself

- For most of the session I could not run code: the shell was blocked until auto mode was switched off in this final turn. Every script was therefore written unexecuted and validated by your runs. One consequence: `test_cleaved_complex.py` still carries an unused `present_atoms()` helper that I reintroduced by mistake.
- One commit message (87dc571) has a stray line that refers to non-existent session notes. The file content is fine; the message cannot be amended without rewriting history.

7 To do

7.1 Recommended next step

Run the control step sweep. Each control was predicted at one step count, and the HA sweep showed that step count can change a fold by more than the effects under test. Crambin’s wrong fold is a property of the protein only if it is wrong at 11, 23 and 46 steps. If it is right at some step count, the crambin calibration has to be redone at that count. The sweep is `run_pdb_comparison.shcontrols`, and its reading was fixed in advance: a fold is correct if the CA RMSD is at most 2.5 Å and the TM-score at least 0.7, and a control is reliable only if correct at every step count of at least $L/4$.

7.2 Prioritised list

1. **Control step sweep** (above).
2. **A local conformation check** for protein G’s loop. pLDDT does not flag residues 9–15, so confidence is not an error check; a local geometry test against experiment is what separates the confident errors.
3. **One verdict table for experiment and prediction**, as in the experimental comparison, stating the fold’s correctness before any distance.
4. **Interface contacts, cleaved against fused**, once the contact metric is validated against experiment on the short proteins.

5. **A verified native disulfide list for HA**, from a named structure entry, asserted at import the way the controls are. Required before any partner-recovery claim about HA.
6. **The trimer: one|two|one|two|one|two**, 1653 positions. The biologically correct question, and expensive.
7. **Condition on a real HA structure** and ask whether ESM3 *maintains* the bridge rather than discovers it. A different and more tractable question.
8. **Only if the cleaved form is needed in the energy machinery**: inject a one-hot at token 31 in the padded region, set the structure token at the break to `STRUCTURE_CHAINBREAK_TOKEN` in `_default()`, and exclude the break site from `mutate()`.

7.3 Housekeeping

- At session end, append the two devlogs to the main one:

```
cat bridge_test/DEVLOG_bridge_test.txt \
    bridge_test/DEVLOG_calibration.txt >> DEVLOG.txt
```

- Give results files run-identifying names; the generic default overwrote the protein G control once.
- Remove the dead `present_atoms()` helper.
- `bridge_test/summarize_runs.py` tabulates every result file from its content; use it instead of copying numbers by hand.

8 Reproducing the runs

From `softmax/code`:

```
python tests/test_cleaved_complex.py --ref protein_g --steps 56
python tests/test_cleaved_complex.py --ref bpti      --steps 58
python tests/test_cleaved_complex.py --ref crambin   --steps 46
python tests/test_cleaved_complex.py --ref cleaved   --steps 256
python tests/test_cleaved_complex.py --ref mature    --steps 256
python tests/test_cleaved_complex.py --ref one       --steps 329
python tests/test_cleaved_complex.py --ref one       --steps 256
python tests/test_cleaved_complex.py --ref two       --steps 221
```

The experimental comparison is `bridge_test/run_pdb_comparison.sh`, with phases `fetch`, `predict`, `compare`, `controls`, `roundtrip` and `prompt`. Output files are in `softmax/code/bridge_test/`.